PROJECT SYNAPSE v1

A single-channel EEG and subvocal-EMG acquisition instrument with variable-length (“Trinx”) gesture decoding

`Analog front-end → 16-bit acquisition → Bluetooth transport → real-time desktop-class decoder`

POWERED BY [mikey:hexoid-sys]

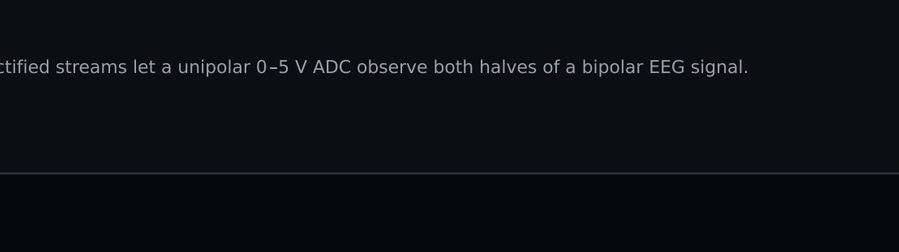
mikey-7x

github.com/mikey-7x/Project-Synapse-v1-official-Real-Time-Brainwave-Decoder-with-Android

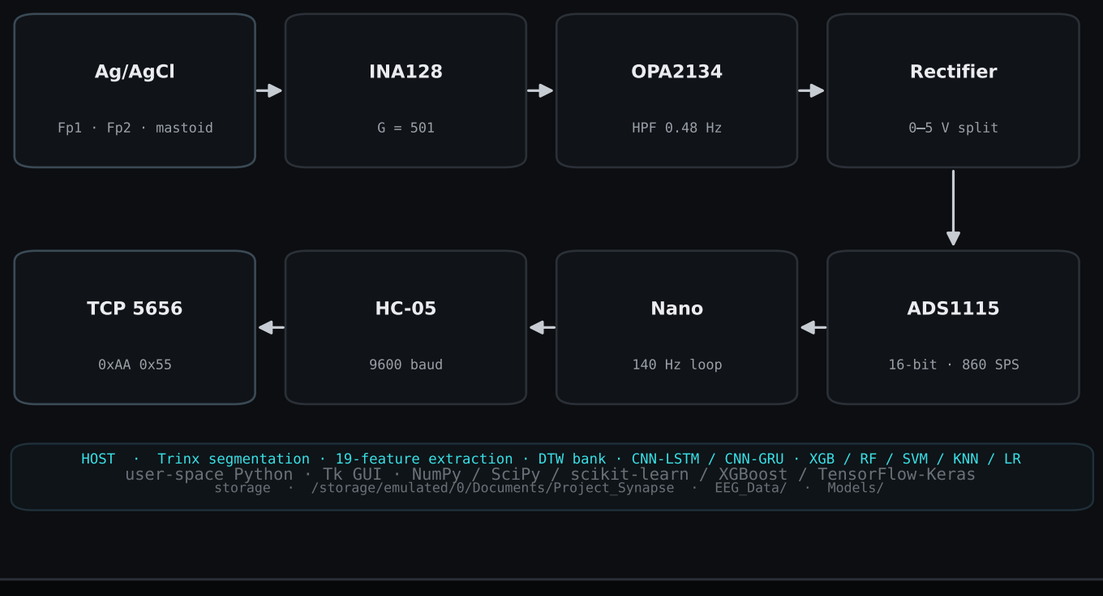
Experimental research prototype · not a medical device

01 The problem: thoughts have no fixed clock

Fixed 2-second windows and zero-padding destroy the shape of a biological gesture: the same subvocalised word may last 0.8 s once and 1.3 s the next. Synapse instead treats a gesture as a time-elastic structure — a Trinx — and matches its topology after amplitude normalisation.



02 System architecture



03 Analog front-end, stage by stage



Dual supply: 3.7 V Li-Po → MT3608 (+5 V) and LM2776 (−5 V). Protection: 1N5819 clamps and RC RFI filter ahead of the INA128.

Two rectified streams let a unipolar 0–5 V ADC observe both halves of a bipolar EEG signal.

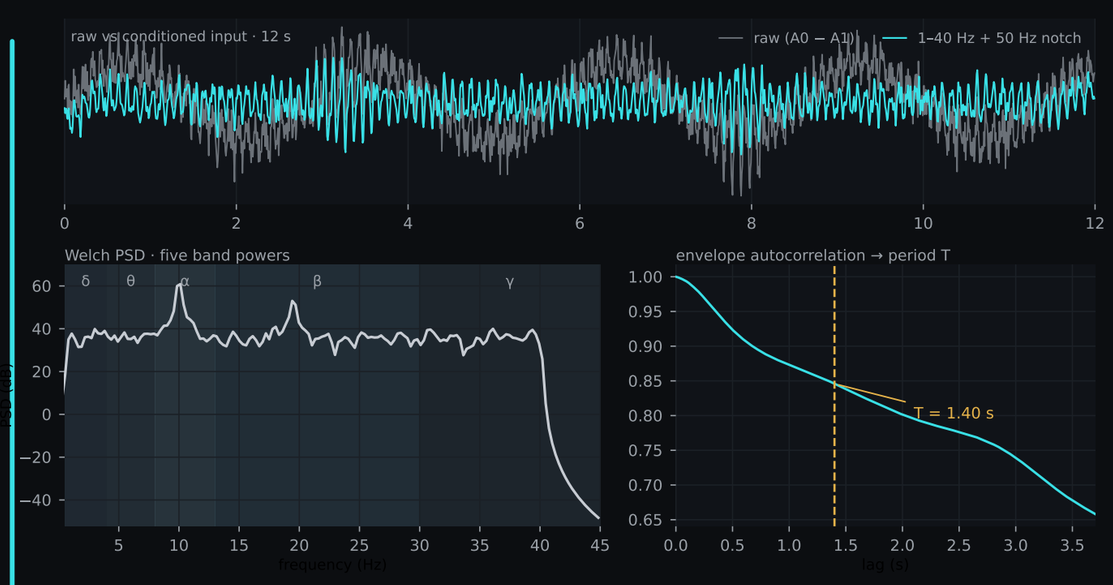
04 Binary link and timing budget



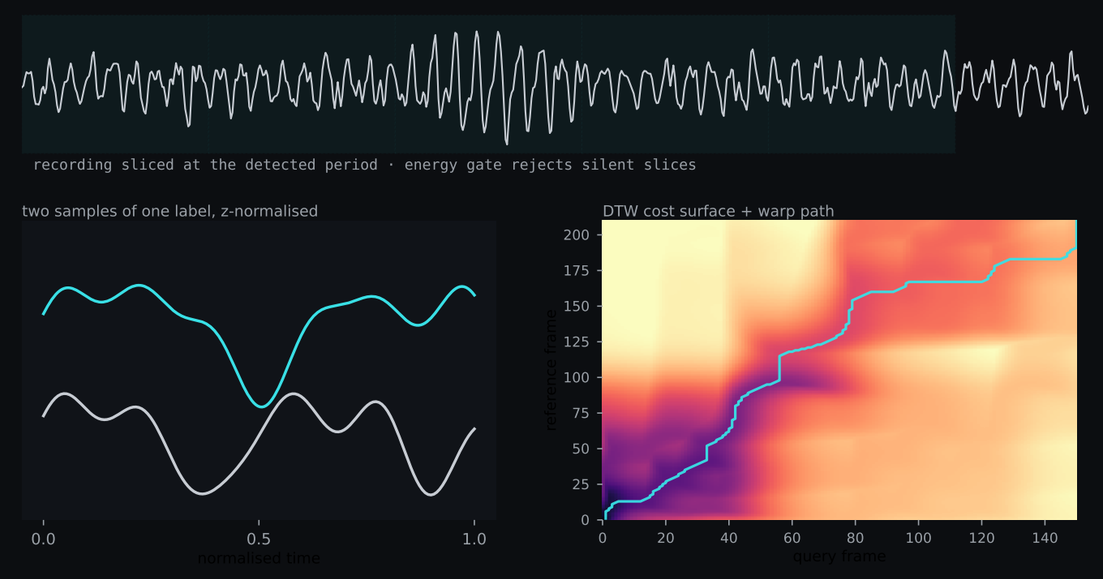
6-byte frame · big-endian · differential channel = A0 − A1 (common-mode rejection in software)

sample rate	**140.0 Hz**	micros()-paced 7 142 µs
ADC rate	**860 SPS**	ADS1115 internal, oversampled
payload	**840 B/s**	140 frames × 6 bytes
link	**9 600 baud**	HC-05 SPP → TCP bridge
UI buffer	**560 samples**	4.0 s rolling window

05 Signal conditioning and spectral view



06 Trinx extraction and elastic matching

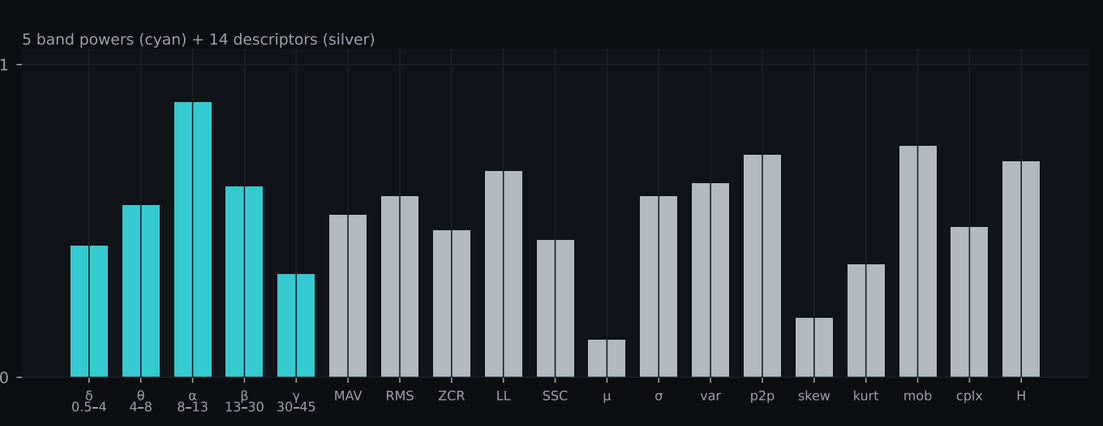


07 Decoder stack

Trinx-DTW	z-normalised DTW memory bank, nearest neighbour
CNN-LSTM	Conv1D(32) → BiLSTM(64) → LSTM(32) → softmax
CNN-GRU	Conv1D(32) → BiGRU(64) → GRU(32) → softmax
Classical	XGBoost · RandomForest · SVM-RBF · KNN · LogReg

Acceptance: a label is printed only at ≥ 90 % confidence; otherwise it is rejected.
Confidence is a distance heuristic, not a calibrated probability.

08 19-value feature vector



LIMITS & SAFETY Not a medical device: no clinical validation, no regulatory certification, no diagnostic claim. Single subject-scale, single-channel data; no published accuracy is claimed here.

BATTERY POWER ONLY while electrodes touch a person — never mains-referenced equipment. Figures on this poster are synthetic illustrations of the pipeline, not measured results.

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